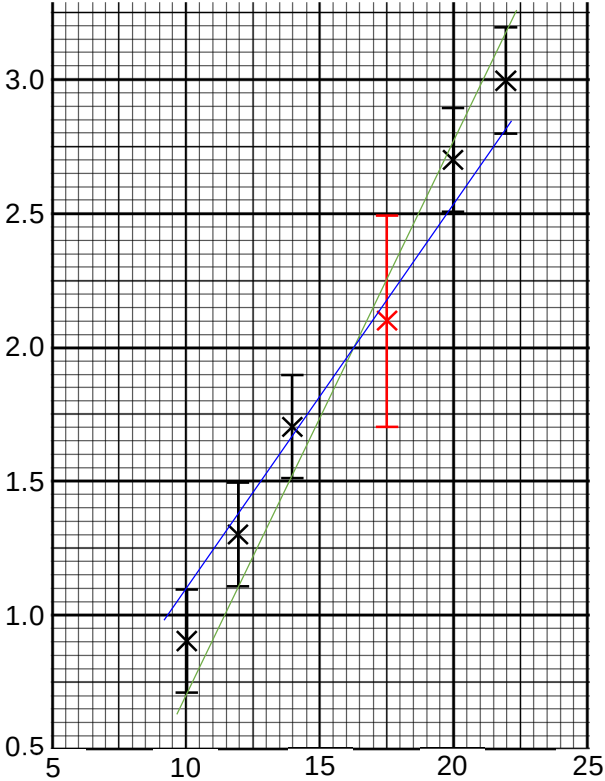


Qns	Answer	Marks
1(a)	same reading so method is valid	B1
	tension acts with <u>same magnitude throughout</u> the elastic band	B1
1(b)(i)	$k = \frac{\text{force}}{\text{extension}} = \frac{T}{x - x_0} = \frac{2.1}{(17.5 - 5) \times 10^{-2}} = 16.8 \text{ N m}^{-1}$	A1
1(b)(ii)	$k = \frac{F}{x - x_0}$ $\frac{\Delta k}{k} = \frac{\Delta T}{T} + \frac{\Delta x + \Delta x_0}{(x - x_0)}$ $\Delta k = k \left(\frac{\Delta T}{T} + \frac{\Delta x + \Delta x_0}{(x - x_0)} \right) = (16.8) \left(\frac{0.4}{2.1} + \frac{0.6}{12.5} \right)$ $= 4.0 \text{ N m}^{-1} \text{ (2 s.f.)}$ or $k = \frac{F}{x - x_0}$ $k_{\max} = \frac{T_{\max}}{(x - x_0)_{\min}} = \frac{2.5}{(17.2 - 5.3) \times 10^{-2}} = 21.0 \text{ N m}^{-1}$ $k_{\min} = \frac{T_{\min}}{(x - x_0)_{\max}} = \frac{1.7}{(17.8 - 4.7) \times 10^{-2}} = 13.0 \text{ N m}^{-1}$ $\Delta k = \frac{1}{2}(k_{\max} - k_{\min}) = \frac{1}{2}(21.0 - 13.0)$ $= \pm 4.0 \text{ N m}^{-1} \text{ (2 s.f.)}$	M1 substitution in correct method A1
1(b)(iii)	$k = 17 \pm 4 \text{ N m}^{-1}$	A1

Qns	Answer	Marks
1(c)		B1 B1 (2 lines)
1(c)(iii)	$\Delta k = \frac{1}{2}(k_{\max} - k_{\min}) = \frac{1}{2} \left(\left. \frac{\Delta y}{\Delta x} \right _{\text{steeper}} - \left. \frac{\Delta y}{\Delta x} \right _{\text{less steep}} \right)$ $= \frac{1}{2} \left(\frac{3.2 - 0.7}{(22 - 10) \times 10^{-2}} - \frac{2.8 - 1.1}{(22 - 10) \times 10^{-2}} \right) = \frac{1}{2} (20.83 - 14.17)$ $= 3.4 \text{ N m}^{-1} \text{ (2 s.f.) (accept 3.3, do not accept 1 s.f.)}$	M1 A0
1(d)	the degree to which the extrapolated data approaches true value of 5 cm is deviating / moving further / moving away, so inaccurate	B1 (contextualize definition of accuracy to question)
2(c)(i)	no	B1